

Introduction

For our project we chose to do an analysis on a data set focused on food inspections in Chicago. The main project question we wanted to focus on was “Is risk level an effective way to measure how ‘risky’ it is to go to a certain restaurant?”

Data Description

The dataset name is "Chicago Department of Public Health (CDPH) Food Inspections." It contains the results of food inspections at different places around Chicago, Illinois. The dataset includes data dating from 2010 up until 2026.

The data was collected by health inspectors who visit not only restaurants but schools, bakeries, grocery stores, etc. There may be many reasons why an inspector may perform an inspection, some of which include license applications, routine inspections, or public complaints. Each row in the dataset represents one inspection.

Now to explain each non-obvious column in the dataset.

Inspection ID is an integer that is used as a unique identifier for each specific inspection.

DBA Name is a string that represents the legal name of an establishment.

AKA Name is another string that represents the informal name of an establishment.

License Number is an integer that is used to identify a specific establishment.

Facility Type is a categorical string that shows what kind of facility the inspection was performed at.

Risk is an Ordinal string that shows the risk level of the facility. Risk level is assigned by how often the restaurant gets inspected.

Inspection Type is a categorical string that shows why the establishment was inspected.

Results is another categorical string that is the outcome of the inspection.

Violations is a string that provides a detailed list of the specific violations that were found.

Cleaning Steps

Before we could get started on the analysis we had to clean the dataset. The date column had to be changed from a string to standard datetime format so we could filter by it later. We filtered the dataset from 2010-2026 down to 2022-2026. This was done to account for changes in food inspection changes along with changes that came with Covid. This also helped with the problem of overplotting as the map was very hard to read with millions of plot points. We also had to convert the Latitude and Longitude columns into numeric types to make sure everything was rendered properly. Finally, for the Violations column we filled empty values with empty strings so we could accurately count the number of violations.

Exploratory Visualizations

When first looking at the dataset we needed to get a hold of trends and what there was to look at. We wanted to explore the location part of the dataset along with the Risk part. Location was very hard to read at first so we decided we needed to filter down the dataset. After filtering it down we realized that many variables had little correlation with location which told us that location has no effect on food inspections in general in Chicago. We then were interested in

Risk Level as we didn't know what it was. We started plotting it against a lot of different variables until we found a few that we thought told a story about the effectiveness of Risk Level to the public.

Visualization Analysis

As soon as I saw the dataset I immediately wanted to know what areas in Chicago had a high density of failed inspections. As I made the visualization I decided that I could expand it into a scatter map that shows specific businesses around Chicago and how they did in recent inspections. I decided to filter the data down from 2010-2026 to 2022-2026 so I could see results of inspections that happened after covid. This visualization can help us answer questions like, are failed inspections a city-wide issue, or a downtown issue. This will help us decide if we want to focus on a specific area code moving forward or not. We ended up noticing that the geography of Chicago had no effect on risk level. This helped us realize we didn't want to lean into the map and area of Chicago any further in this project.

In Visualization #2, we wanted to examine the 'Risk Level' a little more in depth. Looking for a better way to understand it we graphed each risk level in a box plot with 'Number of Violations Found' as the y-axis. We had to create the number of violations found as we had no numerical data in our base dataset. When graphing this we saw a clear correlation between risk level and the number of violations found. Establishments assigned high risk on average committed more violations with medium risk averaging a little less and low averaging the least. Although, since the risk level is assigned based on the amount of times visited, this may just be because the higher risk places get visited more often. To figure out if this was the case we formed another question; Do higher risk places seem to fail inspection more often than low risk places?

Visualization #4 looks at the percentage of passing rates of all establishments depending on their assigned risk level. The word "Risk" made us think that we would see a clear trend of lower passing percentages for the high and medium risk and a high passing percentage for low risk establishments. Surprisingly, we didn't see any of that. Our graph shows a clear lack of correlation between assigned risk level and passing rates. All three assigned risk levels are equally sitting around 60% passing, 24% fail and 17% pass with conditions. Which means there is a reasonable assumption that the differences between each risk level is not statistically significant in terms of passing rates. This does tell us that the trend we saw in Visualization #2 was due to the volume of visits rather than the difference in proportions.

After we graphed Visualization #4 showed that passing rates and risk were not correlated, we were still looking for something that was correlated with Risk Level. We then decided to look into if Risk Level could tell us why these establishments are failing. Visualization #3 is a heatmap that has 'Risk Level' on the x-axis and violation category on the y-axis. The heatmap colors go from light green to dark blue to indicate the percentage of violations. For example, a dark blue equipment box in the high risk column would mean that a lot of high risk establishments do not pass the equipment inspection. Using this, our graph shows an obvious pattern. The high and medium columns have a strong food and physical fail rate. With low risk

having more of a mix between food, physical, toilet, and insect. This tells us that high risk establishments likely failed due to a food or physical violation.

Interactive or geospatial component

As mentioned earlier, we used a map of Chicago in order to show that location has no correlation with Risk Level. The geospatial component of our project was pretty simple in that sense. We decided to use plotly express to plot the map as it offered a clean map with interactive components built in. The interactivity of the map comes from the fact that you can zoom in, out, pan, and click on the key to change which risk levels are visible on the map. Being able to choose what risk level is visible was very useful in not only communicating the visualization, but in analysis as well.

Limitations

We unfortunately felt very limited with this dataset as it lacks a lot of numeric data. Which made us get somewhat creative with how we graphed. In particular, the Violations column had a lot of keywords instead of numerical numbers which meant graphing a distribution would be harder to do. We ended up creating a numerical column for our distribution based visualization. We used the count of total violations of each specific establishment to show the distribution of average violations from high risk, medium risk, and low risk places. We thought this was an effective column that allowed us to communicate that higher risk places have more violations on average.

Conclusions

Through our research we wanted to answer the question: is risk level an effective way to measure how “risky” it is to go to a certain restaurant?

In our geographical distribution we found that there wasn't really any correlation between area and level of risk. There is a clear density of spots downtown but that is consistent across all three risk levels and is most likely explained by there being more shops downtown. Meaning that area alone is not a reliable way to dictate if a place is risky.

Our bar chart had probably the most interesting results of the four. We found that the level of risk assigned to each business had little to no correlation on if it passed inspection. It is possible that there are too many entries in the dataset that failures don't have enough weight to really move the scales on this particular graph but it is something that would be worth looking into.

However, our box plot did show that risk level is correlated with the number of violations per establishment. There was a clear trend with higher risk having more violations and it has a steady trend downwards till low risk. We at first thought that this could show that the ‘riskier’ a place is, the more violations it has. As it's true that a higher risk level means more violations, it's due to the volume of times visited rather than failing more often the lower risk establishments as we figured out in the bar chart visualization.

Finally our heatmap shows a strong correlation with high risk levels and violations with food handling. While low risk establishments are usually failing basic things like infrastructure and cleaning. This suggests that going to a restaurant with a high risk level means that the food handling there likely failed inspection and should be taken into consideration. This was very interesting to see, especially due to our failure to see any trends in earlier visualizations.

What can we make of all this? Well looking at our main question it is clear to assume that risk level is useful but not a perfect system at measuring public safety. While it is true that most high risk establishments failed due to food handling, it does not mean all establishments failed due to food handling. The system lacks nuance that could warn the public of the exact violation to be wary of. We found that Risk Level is assigned by the amount of times a place is visited, to be less misleading we believe that Risk Level should be assigned by total number of violations, or pass/fail rates rather than pure volume. This would be more truthful to the name 'risk' and would help civilians get a better grasp of how risky a place truly is.